

Differential Equations

Lecture 11: Series Solution of Differential Equations

Dr. Shafiq Ur Rehman — UET, Lahore — 2026

Higher Order ODEs: Series Solutions

Motivation and Closed-Form Solutions

In many practically important differential equations the coefficients are not constants but functions of the independent variable x . In such cases, the standard methods for constant-coefficient equations fail to produce a solution expressible as a finite combination of familiar calculus functions (exponentials, polynomials, trig functions, logarithms, etc.). Such a finite combination is called a *closed-form expression*.

Many of the most significant ODEs in applied mathematics – Bessel’s equation, Legendre’s equation, the hypergeometric equation – fall into this category. The standard tool for tackling them is the **power series method**, which seeks solutions written as infinite power series.

Power Series: Review

Definition: Power Series

A **power series centred at x_0** is an infinite series of the form

$$\sum_{k=0}^{\infty} a_k (x - x_0)^k = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots$$

where $a_0, a_1, a_2, \dots$ are constants (the *coefficients*), x_0 is the *centre*, and x is the variable. When $x_0 = 0$ the series becomes

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \cdots$$

Every polynomial is a special case of a power series (with all but finitely many coefficients equal to zero).

Standard Maclaurin Series (to be used freely in examples)

The following series are assumed known:

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1 \quad (\text{Geometric series})$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Ordinary and Singular Points

Consider the second-order linear homogeneous ODE with variable coefficients:

$$a_0(x) y'' + a_1(x) y' + a_2(x) y = 0, \quad a_0(x) \neq 0. \quad (17)$$

Dividing through by $a_0(x)$ gives the **standard form**:

$$y'' + P(x) y' + Q(x) y = 0, \quad (18)$$

where $P(x) = \frac{a_1(x)}{a_0(x)}$ and $Q(x) = \frac{a_2(x)}{a_0(x)}$.

Definitions: Ordinary Point and Singular Point

Let $x = x_0$ be a point in an open interval I .

- x_0 is called an **ordinary point** (or *regular point*) of equation (18) if $a_0(x_0) \neq 0$ and both $P(x)$ and $Q(x)$ are analytic at x_0 .
- x_0 is called a **singular point** of equation (18) if it is *not* an ordinary point, i.e., if $a_0(x_0) = 0$ or if $P(x)$ or $Q(x)$ fails to be analytic at x_0 .

Example: Finding Ordinary and Singular Points of $x^3(x-2)^2 y'' + 5(x+2)(x-2)y' + 3x^2 y = 0$

Classification: This is a second-order linear ODE with variable coefficients. We identify a_0 , a_1 , a_2 then form P and Q .

Step 1 – Identify the coefficient functions.

Comparing with equation (17):

$$a_0(x) = x^3(x-2)^2, \quad a_1(x) = 5(x+2)(x-2), \quad a_2(x) = 3x^2.$$

Step 2 – Find where $a_0(x) = 0$.

$$x^3(x-2)^2 = 0 \implies x = 0 \text{ and } x = 2.$$

These are candidates for singular points.

Step 3 – Form $P(x)$ and $Q(x)$ and check analyticity.

$$P(x) = \frac{a_1(x)}{a_0(x)} = \frac{5(x+2)(x-2)}{x^3(x-2)^2} = \frac{5(x+2)}{x^3(x-2)},$$

$$Q(x) = \frac{a_2(x)}{a_0(x)} = \frac{3x^2}{x^3(x-2)^2} = \frac{3}{x(x-2)^2}.$$

Both $P(x)$ and $Q(x)$ have denominators that vanish at $x = 0$ and $x = 2$, so neither function is analytic at these points.

Conclusion.

$x = 0$ and $x = 2$ are singular points; all other real x are ordinary points.

The Power Series Method

The power series method finds a solution to

$$a_0(x)y'' + a_1(x)y' + a_2(x)y = 0 \quad (19)$$

about an *ordinary* point $x = x_0$ by assuming $y(x)$ has a power series representation and determining the unknown coefficients.

Method: Power Series Solution About an Ordinary Point x_0

Step 1 – Assume a series solution.

$$y(x) = \sum_{k=0}^{\infty} a_k(x-x_0)^k = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + \cdots \quad (20)$$

Step 2 – Differentiate term by term.

$$y'(x) = \sum_{k=1}^{\infty} k a_k(x-x_0)^{k-1} = a_1 + 2a_2(x-x_0) + 3a_3(x-x_0)^2 + \cdots \quad (21)$$

$$y''(x) = \sum_{k=2}^{\infty} k(k-1) a_k(x-x_0)^{k-2} = 2a_2 + 6a_3(x-x_0) + 12a_4(x-x_0)^2 + \cdots \quad (22)$$

Note: the series for y' starts at $k = 1$ and for y'' at $k = 2$.

Step 3 – Substitute into the ODE.

Insert y , y' , y'' into equation (19). Collect terms by powers of $(x-x_0)$ and set the coefficient of each power equal to zero. This produces a *recurrence relation* linking the a_k .

Step 4 – Write the general series solution.

Express all a_k in terms of the two arbitrary constants a_0 and a_1 , then substitute back into equation (20).

Note: The series for y' and y'' begin at different indices ($k = 1$ and $k = 2$ respectively). When substituting into the ODE, all series must be aligned to the same power of $(x-x_0)$ before collecting terms – shift the index of summation as needed.

Worked Examples

Example 1: Power Series Solution of $y'' + y = 0$ about $x_0 = 0$

Classification: Constant-coefficient ODE; $x_0 = 0$ is an ordinary point (the coefficients 1 and 1 are analytic everywhere). We use the power series method to recover the familiar $\sin x$ and $\cos x$ solution.

Step 1 – Assume the series solution about $x_0 = 0$.

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots \quad (23)$$

Step 2 – Compute the derivatives.

$$y'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \dots$$

$$y''(x) = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \dots$$

Step 3 – Substitute into $y'' + y = 0$ and equate coefficients to zero.

Adding y'' and y :

$$(2a_2 + a_0) + (6a_3 + a_1)x + (12a_4 + a_2)x^2 + (20a_5 + a_3)x^3 + \dots = 0.$$

Setting each coefficient to zero:

$$x^0: \quad 2a_2 + a_0 = 0 \quad \implies \quad a_2 = -\frac{a_0}{2} = -\frac{a_0}{2!}$$

$$x^1: \quad 6a_3 + a_1 = 0 \quad \implies \quad a_3 = -\frac{a_1}{6} = -\frac{a_1}{3!}$$

$$x^2: \quad 12a_4 + a_2 = 0 \quad \implies \quad a_4 = -\frac{a_2}{12} = \frac{a_0}{24} = \frac{a_0}{4!}$$

$$x^3: \quad 20a_5 + a_3 = 0 \quad \implies \quad a_5 = -\frac{a_3}{20} = \frac{a_1}{120} = \frac{a_1}{5!}$$

Here a_0 and a_1 remain *arbitrary* (two free constants for a second-order ODE).

Step 4 – Substitute back into (23) and rearrange.

$$\begin{aligned} y(x) &= a_0 + a_1x - \frac{a_0}{2!}x^2 - \frac{a_1}{3!}x^3 + \frac{a_0}{4!}x^4 + \frac{a_1}{5!}x^5 - \dots \\ &= a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right). \end{aligned}$$

Recognising the standard Maclaurin series:

$$y(x) = a_0 \cos x + a_1 \sin x.$$

This confirms the familiar general solution – the power series method reproduces the exact closed form in this case.

Example 2: Power Series Solution of $(1 - x^2)y'' - 2xy' + 2y = 0$ about $x_0 = 0$

Classification: Variable-coefficient ODE. Writing in standard form: $P(x) = \frac{-2x}{1 - x^2}$, $Q(x) = \frac{2}{1 - x^2}$. Both are analytic at $x_0 = 0$ (the only singularities are at $x = \pm 1$), so $x_0 = 0$ is an ordinary point.

Step 1 – Assume the series.

$$y(x) = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

Step 2 – Differentiate.

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \dots$$

Step 3 – Substitute into $(1 - x^2)y'' - 2xy' + 2y = 0$.

Expanding each term:

$$(1 - x^2)y'' = 2a_2 + 6a_3x + (12a_4 - 2a_2)x^2 + (20a_5 - 6a_3)x^3 + \dots$$

$$-2xy' = -2a_1x - 4a_2x^2 - 6a_3x^3 - \dots$$

$$2y = 2a_0 + 2a_1x + 2a_2x^2 + 2a_3x^3 + \dots$$

Adding and setting each coefficient to zero:

$$x^0: 2a_2 + 2a_0 = 0 \implies a_2 = -a_0$$

$$x^1: 6a_3 - 2a_1 + 2a_1 = 0 \implies a_3 = 0$$

$$x^2: 12a_4 - 2a_2 - 4a_2 + 2a_2 = 0 \implies a_4 = \frac{4a_2}{12} = \frac{4(-a_0)}{12} = -\frac{a_0}{3}$$

$$x^3: 20a_5 - 6a_3 - 6a_3 + 2a_3 = 0 \implies a_5 = 0$$

The pattern continues with all odd-indexed coefficients (beyond a_1) vanishing.**Step 4 – General solution.**

$$y(x) = a_0 \left(1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 - \dots \right) + a_1 x.$$

The second solution is just $y_2 = x$ (a polynomial), while the first series converges for $|x| < 1$.

Key observation: In Example 2, the second linearly independent solution is simply $y_2(x) = x$ – a polynomial. This happens precisely because one coefficient in the recurrence eventually forces a terminating series. Such terminating power-series solutions are called *polynomial solutions*; Legendre's equation produces them as Legendre polynomials.

Systems of Linear Differential Equations

Introduction

So far we have studied ODEs with a single dependent variable y and one independent variable x . Many physical problems, however, involve *several* dependent variables and a single independent variable. The resulting coupled equations form a **system of simultaneous linear differential equations**.

A complete solution can be obtained only when the number of equations equals the number of dependent variables.

General Form: Two First-Order Equations

With dependent variables $x(t)$ and $y(t)$ and independent variable t , the simplest system is:

$$\begin{aligned}\frac{dx}{dt} &= a_{11}x + a_{12}y + f_1(t) &\iff (D - a_{11})x - a_{12}y &= f_1(t) \\ \frac{dy}{dt} &= a_{21}x + a_{22}y + f_2(t) &\iff -a_{21}x + (D - a_{22})y &= f_2(t)\end{aligned}$$

where $D = \frac{d}{dt}$, and all a_{ij} are constants. If $f_1 = f_2 = 0$ the system is *homogeneous*; otherwise it is *non-homogeneous*.

Method: Elimination Method for 2-Variable Systems

Step 1. Rewrite the system using the operator $D = d/dt$.

Step 2. Operate on one equation to eliminate one dependent variable (say y). This leaves a single higher-order ODE in x alone.

Step 3. Solve the resulting ODE for $x(t)$ using standard methods (characteristic equation, etc.).

Step 4. Substitute the solution $x(t)$ back into one of the original equations to obtain $y(t)$ by differentiation – no further integration is needed.

Step 5. Write the general solution as the pair $\{x(t), y(t)\}$.

Example 3: Solving the System $\dot{x} = y, \dot{y} = -4x + 4y$

Classification: Homogeneous system with constant coefficients; solved by the elimination method.

Step 1 – Operator form.

$$Dx - y = 0 \tag{24}$$

$$4x + (D - 4)y = 0 \tag{25}$$

Step 2 – Eliminate y .

Apply operator $(D - 4)$ to equation (24):

$$(D - 4)(Dx - y) = 0 \implies D^2x - 4Dx - Dy + 4y = 0. \tag{*}$$

From (24): $Dy = D(Dx) = D^2x$ (differentiating $Dx = y$ gives $D^2x = Dy$). Adding (*) and (25):

$$D^2x - 4Dx + 4x = 0 \implies (D^2 - 4D + 4)x = 0.$$

Step 3 – Solve for $x(t)$.

Auxiliary equation: $m^2 - 4m + 4 = (m - 2)^2 = 0$, giving repeated roots $m = 2, 2$.

Since the roots are real and repeated:

$$x(t) = (c_1 + c_2t) e^{2t}.$$

Step 4 – Recover $y(t)$.

From equation (24), $y = Dx$:

$$\begin{aligned}y &= \frac{d}{dt} [(c_1 + c_2t) e^{2t}] \\ &= c_2 e^{2t} + 2(c_1 + c_2t) e^{2t} \\ &= 2c_1 e^{2t} + c_2 e^{2t} + 2c_2t e^{2t}.\end{aligned}$$

General solution.

$$\boxed{x(t) = (c_1 + c_2t) e^{2t}, \quad y(t) = (2c_1 + c_2 + 2c_2t) e^{2t}.$$

Common mistake: After finding $x(t)$, students sometimes try to integrate an equation to get $y(t)$. It is always cleaner – and avoids introducing an extra arbitrary constant – to use a *differentiation* step (as in Step 4 above) rather than integration.

Summary Table

Topic	Key Condition / Standard Form	Solution Approach
Ordinary point	$a_0(x_0) \neq 0$; P, Q analytic at x_0	Power series about x_0
Singular point	$a_0(x_0) = 0$ or P/Q not analytic	Frobenius method (not covered here)
Power series method	$y = \sum a_k(x - x_0)^k$; substitute and equate	Recurrence relation $\rightarrow$ coefficients in terms of a_0, a_1
Systems of ODEs	$(D - a_{11})x - a_{12}y = f_1(t)$ etc.	Operator elimination $\rightarrow$ higher-order ODE in one variable

Exercise (Ref.: Book 1, Problem Set 5.1, Questions 10–14). Solve the following using the power series method about $x_0 = 0$:

- (1) $y'' - xy' - y = 0$
- (2) $(1 + x^2)y'' - 4xy' + 6y = 0$

Exercise (Ref.: Book 1, Problem Set 4.6, Questions 1–15). Solve the following systems using the elimination method.